

Think Twice: Does Repeating Reasoning Summaries Increase Chain-of-Thought Efficiency?

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Abstract:

Reasoning-capable language models often improve final-answer accuracy by generating long chains of thought, but this improvement can come with substantial inference cost. At the same time, two recent lines of work suggest that the form and repetition of context can strongly affect model behavior. Repeating the prompt, by duplicating the user query at least once, on the one side shows improvements in non-reasoning language model performance, while on the other side repeating the raw reasoning tokens of one request into another, similar request can degrade reasoning models. Maneuvering around this pitfall the approach of Insights-to-Solver (I2S) was proposed, where distilled reasoning summaries can improve performance of reasoning models. This paper studies the unexplored intersection of these findings: whether repeating a distilled reasoning summary improves answer accuracy, reduces generated reasoning tokens, or both.

We propose a controlled evaluation of extracting an abstract, task-relevant reasoning insight from a solved exemplar and inserting the same summary before a new problem with different repetition counts $k \in \{1, 2, 3, 4, 5\}$. We compare the results against direct prompting, pure prompt repetition, and raw reasoning-transfer baselines at matched insight repetition. The study is designed for both reasoning and non-reasoning settings, using Qwen3-4B with thinking enabled and disabled as the primary controlled comparison, plus a second model family for robustness. — Results —

Keywords: machine learning, reasoning, context scaling, chain-of-thought

Project Page: https://github.com/Archerui/llm_precomputed_context_exp

1 Introduction

Large language models (LLMs) can solve difficult mathematical, symbolic, and commonsense tasks more reliably when they produce intermediate reasoning. Chain-of-thought prompting and self-consistency decoding made this phenomenon visible for general-purpose models, and recent large reasoning models extend it by spending more test-time compute on explicit or hidden reasoning. However, longer reasoning is not free. It increases generated tokens, latency, and cost, and it can produce verbose traces that are difficult to audit. A central question for efficient reasoning is therefore not only whether models can think more, but whether they can think more selectively Didolkar et al..

1.1 Problem and Motivation

Two recent lines of observation create a useful tension for addressing this efficiency problem. First, prompt repetition can improve non-reasoning language models by simply duplicating the input query Leviathan et al. (2025).

Second, research on test-time demonstrations shows that transferring raw reasoning traces as few-shot examples can actually hurt reasoning models through "semantic misguidance," whereas extracting concise, abstract insights or "behaviors" from those traces restores and improves performance Wang et al. (2025), ?. The first result suggests that repetition alone alters model attention paths mechanically; the second suggests that distilled procedural memory is semantically safer and more efficient than raw traces.

What remains unclear is what happens when the repeated object is not the original prompt, but an abstract reasoning summary.

1.2 The Unexplored Gap: Semantic Value vs. Mechanical Load

While repeating a reasoning summary seems intuitively promising, the interaction between the mechanical gains of repetition and the semantic injection of abstract reasoning remains entirely unexplored. Recent mechanis-

tic studies reveal that inserting long sequences of even completely meaningless tokens induces a beneficial "activation redistribution" in the model's multi-layer perceptrons (MLPs), pushing near-zero activations outward and enhancing the model's representational capacity Shi et al. (2025). This raises a critical question: if repeating a summary improves performance, is the gain purely *semantic* (driven by the abstract procedural rules), or is it merely *mechanical* (acting as an affine transformation on attention outputs due to the increased token length)?

Furthermore, naive repetition inherently increases the context length. Drawing on Cognitive Load Theory, recent architectural studies like REPO show that prevailing LLMs suffer from "extraneous cognitive load" due to their rigid, linear positional encodings. Bloating the prompt with repeated text consumes finite working memory capacity that should be allocated to deep reasoning, potentially impairing performance. Therefore, it is unknown whether the bidirectional attention gains of repeating a summary can stack with its semantic value, or if the increased prompt length will over-constrain the model and induce cognitive overload.

1.3 Solution Approach

To untangle these effects, we propose a controlled evaluation of *summary repetition*. We extract an abstract, task-relevant reasoning insight from a solved exemplar (aligning with procedural behaviors) and insert the exact same summary before a new problem with differing summary repetition counts $k \in \{1, 2, 3, 4, 5\}$. Crucially, to isolate whether performance gains are semantic or mechanical, we compare summary repetition (SR_k) against direct prompting, pure prompt repetition (PR_k), and raw reasoning-transfer baselines (RT_k) at matched copy counts.

If repeated summaries improve accuracy or reduce the amount of generated reasoning tokens, they offer a highly token-efficient, inference-time method for guiding a model without fine-tuning. Conversely, if they fail, it demonstrates that the mechanical benefits of prompt repetition do not easily transfer to semantic reasoning content, or that the extraneous cognitive load of repeated summaries outweighs their procedural guidance.

1.4 Contributions

The planned study makes four contributions:

- First, it formalizes as the intersection of prompt repetition and distilled reasoning transfer.
- Second, it introduces a copy-count-matched experimental design comparing summary repetition, pure prompt repetition, and raw reasoning transfer.
- Third, it evaluates both accuracy and token efficiency, including generated-token counts rather than only visible or hidden reasoning tokens.
- Fourth, it includes a second model family to distinguish Qwen-specific behavior from more general effects.

2 Related Work

2.1 Prompt Repetition & Context Scaling

Prompt repetition studies the surprisingly simple intervention of duplicating the input prompt. Prompt repetition can improve non-reasoning language models across several benchmarks, and attribute the effect to how duplicate prompt tokens alter attention paths in causal transformers. This line of work is related to broader evidence that language models are sensitive to context order and position. However, prompt repetition repeats the problem statement itself. It does not test whether a repeated abstract hint, skill, or summary can have a similar or stronger effect.

2.2 Reasoning Summaries and Traces

Chain-of-thought prompting showed that language models can improve multi-step reasoning when prompted to generate intermediate natural-language steps. Self-consistency further improved performance by sampling multiple reasoning paths and marginalizing over final answers. Tree-of-thought methods generalized this idea by allowing explicit search over intermediate states. Few-shot reasoning demonstrations can help models by showing examples of intermediate reasoning, but raw traces are not always beneficial. I2S directly studies this issue for reasoning models and reports that raw reasoning traces can degrade performance, while abstracted insight summaries extracted from those traces can improve it. This motivates our choice of repeated object: we repeat the distilled summary rather than the full trace. The key difference from I2S is that we treat the number of summary copies as an experimental variable.

Several recent methods try to reuse reasoning more efficiently. TRS extracts and reuses reasoning skills, showing that reusable skills can improve reasoning without always requiring long demonstrations. Retrieval-of-Thought (RoT) investigates retrieving useful thoughts for new problems, aiming to combine retrieval with reasoning-time guidance. ThoughtMani studies manipulating a model's reasoning process with more compact interventions. CoThink uses collaboration among models to improve reasoning and reduce unnecessary thinking. These methods share the intuition that reasoning can be transferred, compressed, retrieved, or coordinated.

3 Method

3.1 Task Formulation

3.2 Architecture Design & Context Positioning

3.3 Training & Evaluation Objective

We define a simple objective:

$$\mathcal{L} = E_{x \sim \mathcal{D}} [\ell(f_\theta(x), y)] \quad (1)$$

Table 1: This is a detailed table description. You must be able to understand the content purely based on this description and the table content, without reading the full text.

Heading (a)	Group 1			Group 2		
	M_{1_a}	M_{1_b}	M_2	M_{1_a}	M_{1_b}	M_2
<i>Some additional information</i>						
C_1 (Description/Venue)	0.9830	-0.0003	0.9558	0.5480	0.0781	0.8687
C_2 (Description/Venue)	0.9894	0.0001	0.8756	[-]	[-]	[-]
C_3 (Description/Venue)	1.0106	0.0000	0.0420	[-]	[-]	[-]

4 Experimental Apparatus

4.1 Datasets

4.2 Preprocessing

4.3 Procedure & Baselines

4.4 Metrics

4.5 Main Results

4.6 Ablation Study

4.7 Qualitative Results

5 Discussion

5.1 Limitations

5.2 Broader Impact

6 Conclusion

References

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A Additional Details

A.1 Extended Results

A.2 Implementation Details